

Case Study: University Management System (Hibernate Mappings)

Business Background

A university wants to develop a backend system to manage:

- Students
- Courses
- Departments
- Student ID Cards

The system must maintain relationships between these entities using **Hibernate ORM**.

Your task is to implement the data model and demonstrate all mappings.

Objective

Create a Hibernate project that demonstrates:

Mapping Type	Where It Appears
One-to-One	Student ↔ IDCard
One-to-Many	Department → Students
Many-to-One	Student → Department
Many-to-Many	Student ↔ Course

Entity Relationship Requirements

One-to-One

Student ↔ IDCard

- Each student has one ID card.
 - Each ID card belongs to exactly one student.
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One-to-Many

Department → Students

- One department has many students.
 - One student belongs to only one department.
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Many-to-Many

Student ↔ Course

- A student can enroll in multiple courses.
 - A course can have multiple students.
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Entities to Create

Department

Fields:

- id
- Name

Relationships:

- `@OneToMany(mappedBy = "department")`
 - `List<Student>`
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Student

Fields:

- `id`
- `name`

Relationships:

- `@ManyToOne` → Department
 - `@OneToOne` → IDCard
 - `@ManyToMany` → Course
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IDCard

Fields:

- `id`
- `cardNumber`

Relationship:

- `@OneToOne(mappedBy = "idCard")`
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Course

Fields:

- `id`
- `courseName`

Relationship:

- `@ManyToMany(mappedBy = "courses")`
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Tasks

Task 1: Configure Hibernate

Create:

- `hibernate.cfg.xml`
 - Add all entity mappings
 - Enable `show_sql`
 - Use `hbm2ddl.auto=update`
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Task 2: Implement All Mappings Using Annotations

You must:

- Use `@JoinColumn`
 - Use `@JoinTable` for Many-to-Many
 - Define owning side correctly
 - Maintain `mappedBy` where required
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Task 3: Perform Operations in Main Class

In `Main.java`:

1. Create a Department
2. Create two Students
3. Assign both students to the same Department
4. Create ID cards for each student

5. Create two Courses
 6. Enroll students in courses (Many-to-Many)
 7. Save everything using proper transactions
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Expected Database Structure

Hibernate should generate:

- department table
 - student table
 - idcard table
 - course table
 - student_course (join table)
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